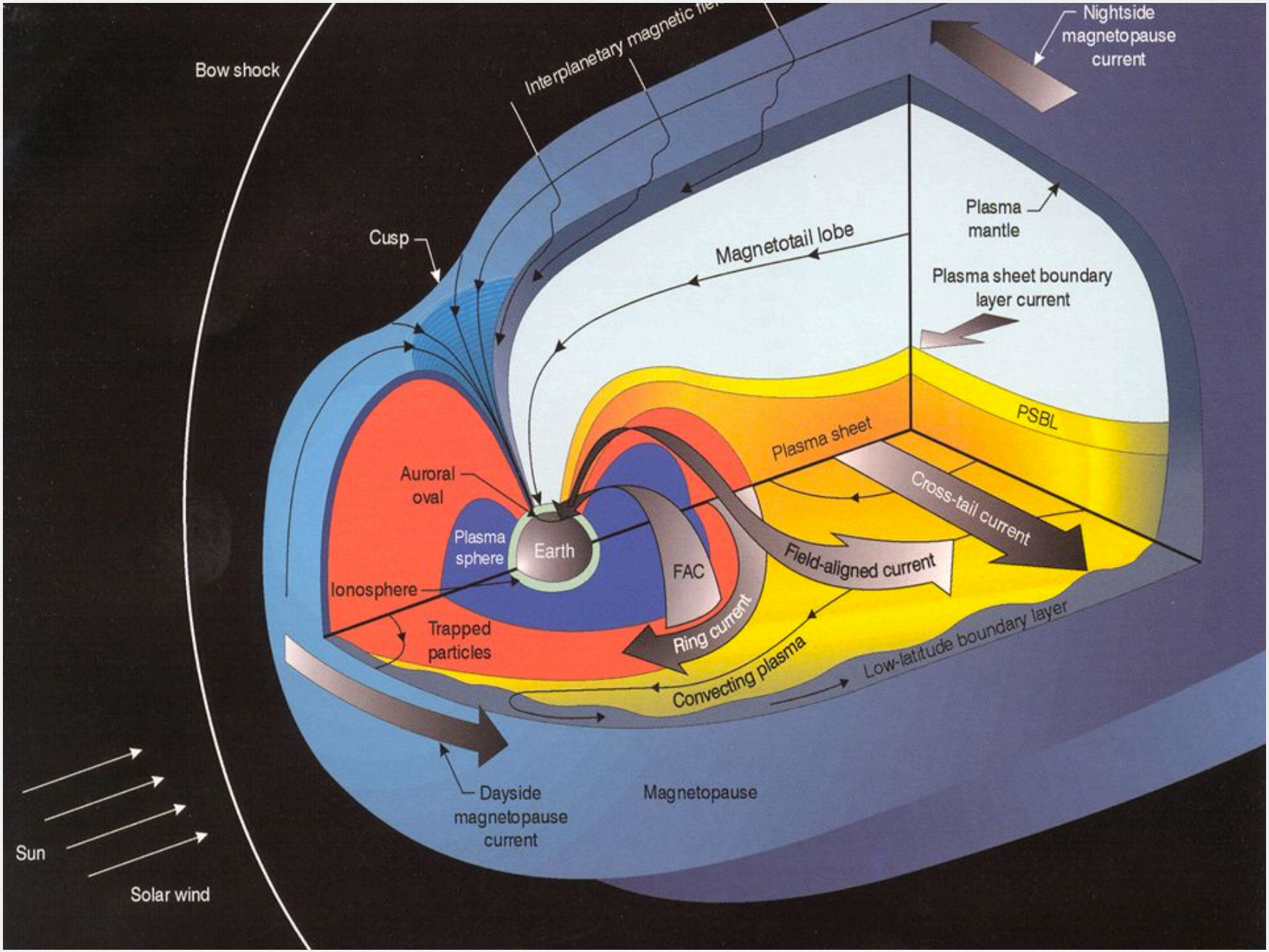


University of Colorado **Boulder**

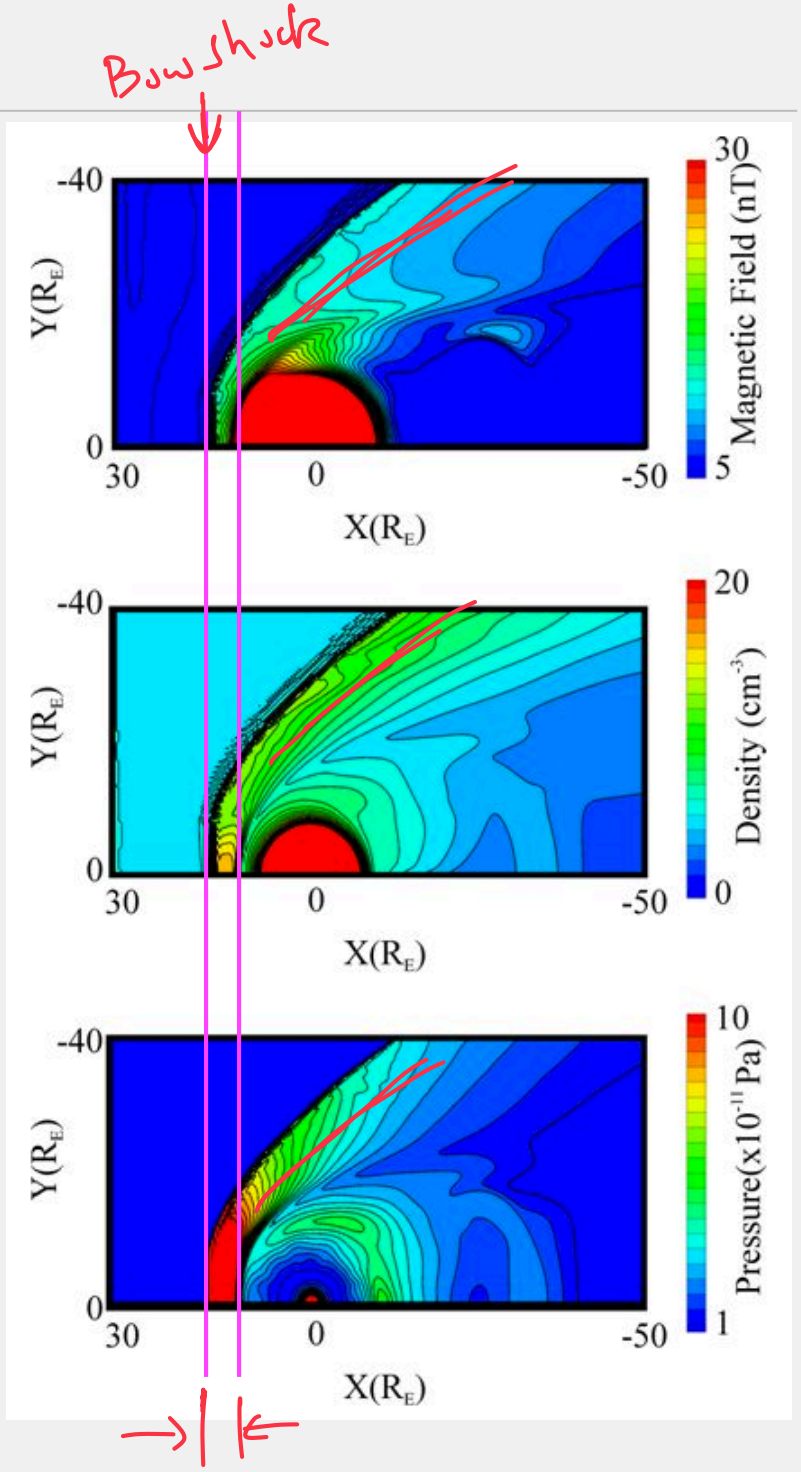
Particles and Currents in the Magnetosphere

Basic Structure of (Inner) Magnetosphere



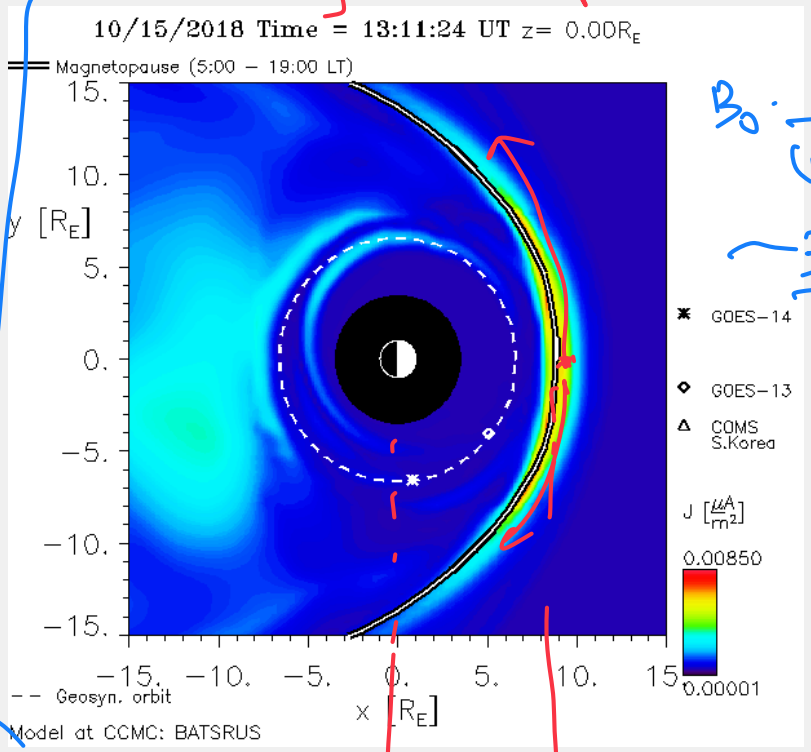
Bow Shock

- Occurs where solar wind goes from supersonic to subsonic
  - 3-4 Earth radii in front of the magnetopause
  - Ranges in thickness from a few to 100s of km
  - Solar wind is slowed, heated, and compressed
  - Between bow shock and magnetopause: **magnetosheath**; contains this compressed, heated solar wind plasma
  - Different from typical shocks (such as an airplane) because it is i) collisionless, and ii) stationary



Magnetopause

- Occurs where solar wind **dynamic pressure** equals **magnetic pressure** (from magnetosphere)
- solar wind essentially *stops*; forced to flow around the magnetosphere **(why?)**
- Some solar wind plasma manages to enter the magnetosphere, becoming trapped **(wait, why?)** becoming the plasma sheet (later)
- **Standoff distance** (justification for the following equation?)**:**



$B_0 \sim 31{,}000\,nT$

$B_{mag} = B_0 \left(\frac{R_E}{r}\right)^3 = B_0 \cdot \frac{1}{L^3}$

$B_0 \cdot \frac{1}{(10)^3} \sim 31\,nT$

$\frac{B_{sw}^2}{2\mu_0} + \rho_{sw} v_{sw}^2 = \frac{(2B_{mag})^2}{2\mu_0} + \frac{f}{2}\rho_{mag} \cdot v_{th}^2$

$= \rho_{sw} v_{sw}^2 = \frac{4B_0^2}{2\mu_0 L^6} \Rightarrow L = 10$

$\rho_{sw} \sim 5\ ion/cm^3 \quad (m_p)$

$v_{sw} \sim 400\,km/s$

$B \sim 5\,nT$

Magnetopause current, aka, Chapman-Ferraro Current (1930)

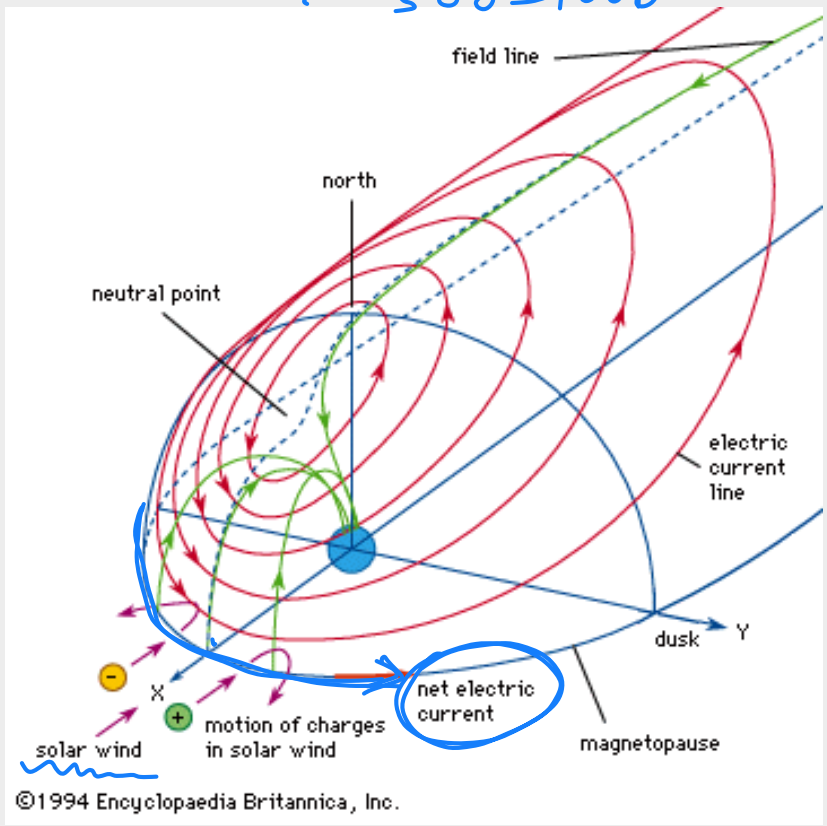
looking down from North

B_{mag} 30 nT at L = 10

– MP thickness

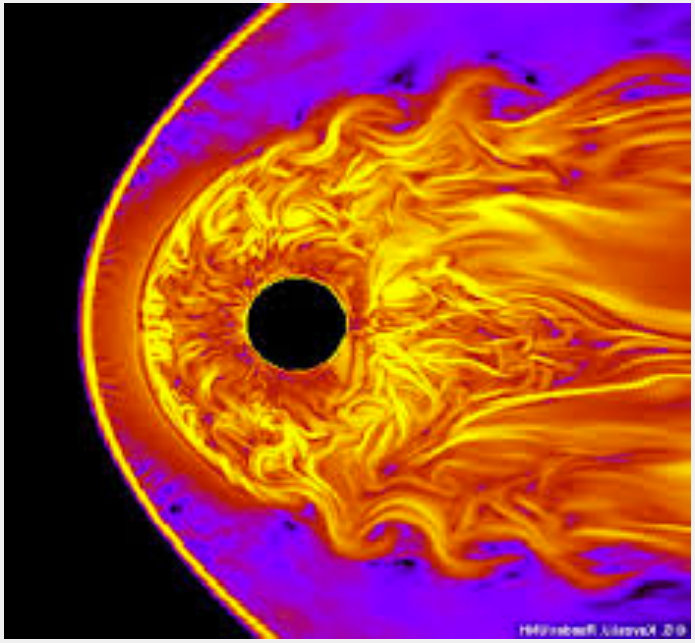
~ $\frac{1}{5}$ $r_{g,proton}$

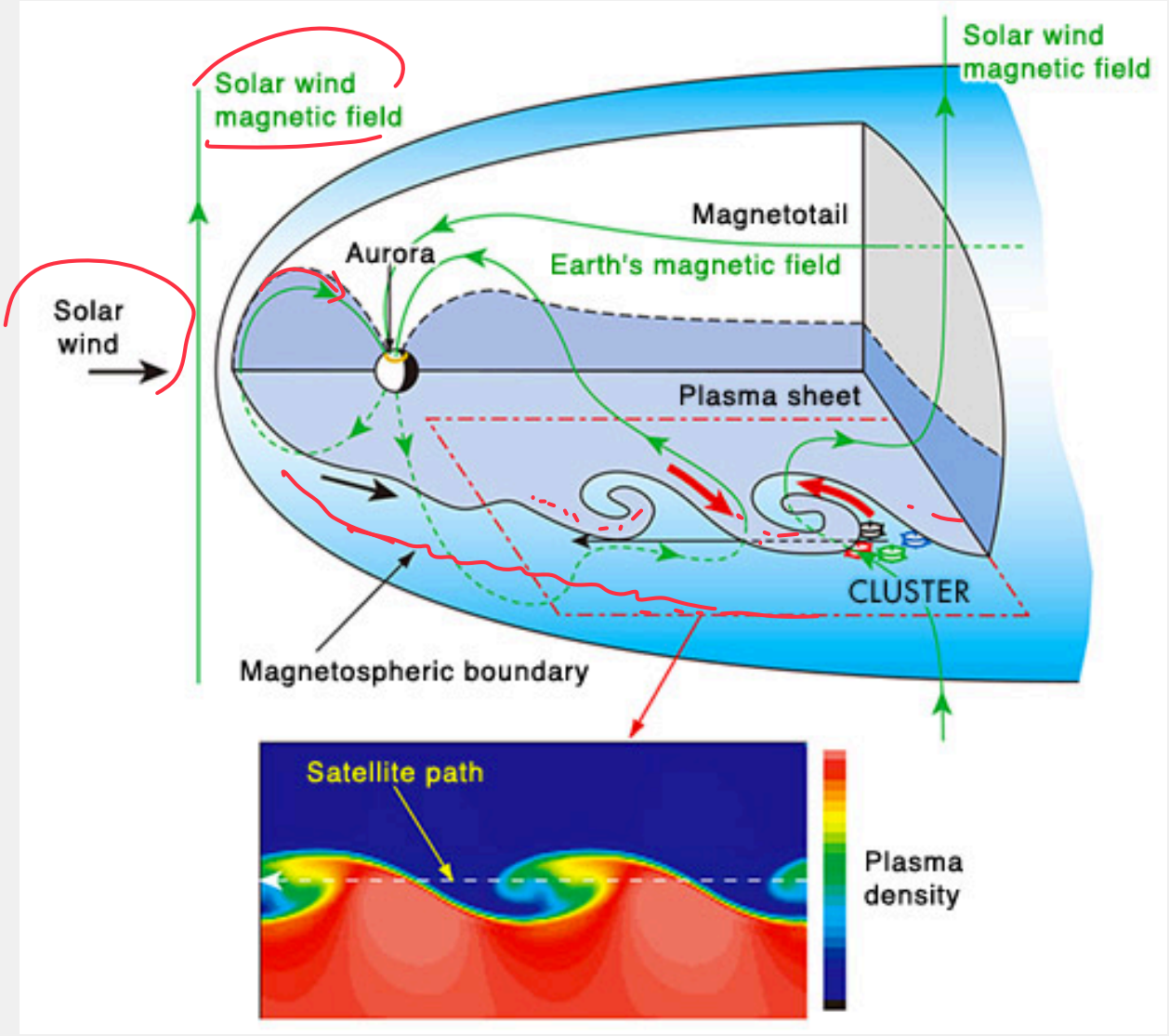
~ 500–1000 km



Wiggles in the Magnetopause

- As solar wind flows around the magnetosphere in the **magnetosheath**, there is a shear flow at the magnetopause
  - Drives Kelvin-Helmholtz instability (more significant, relatively speaking, during northward IMF)



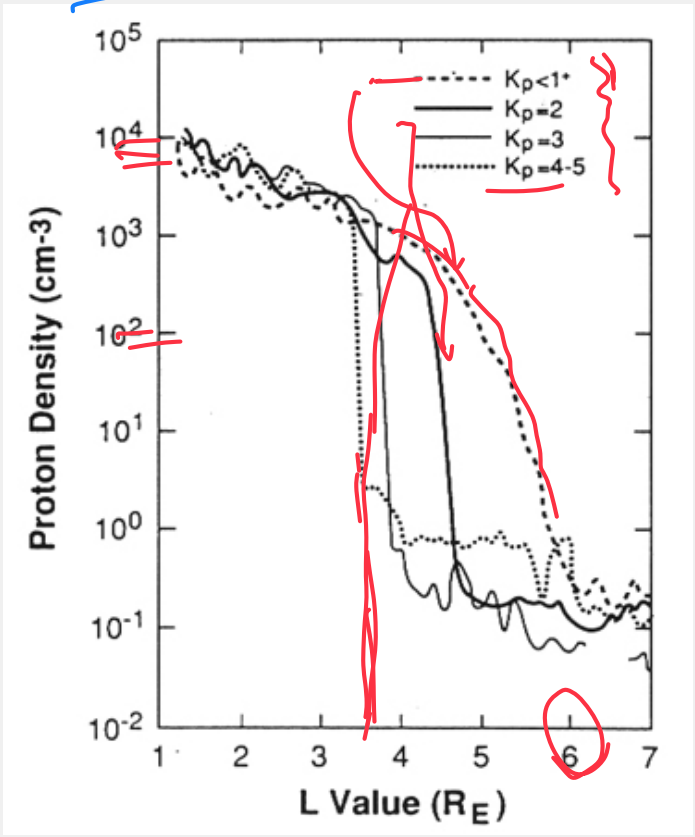


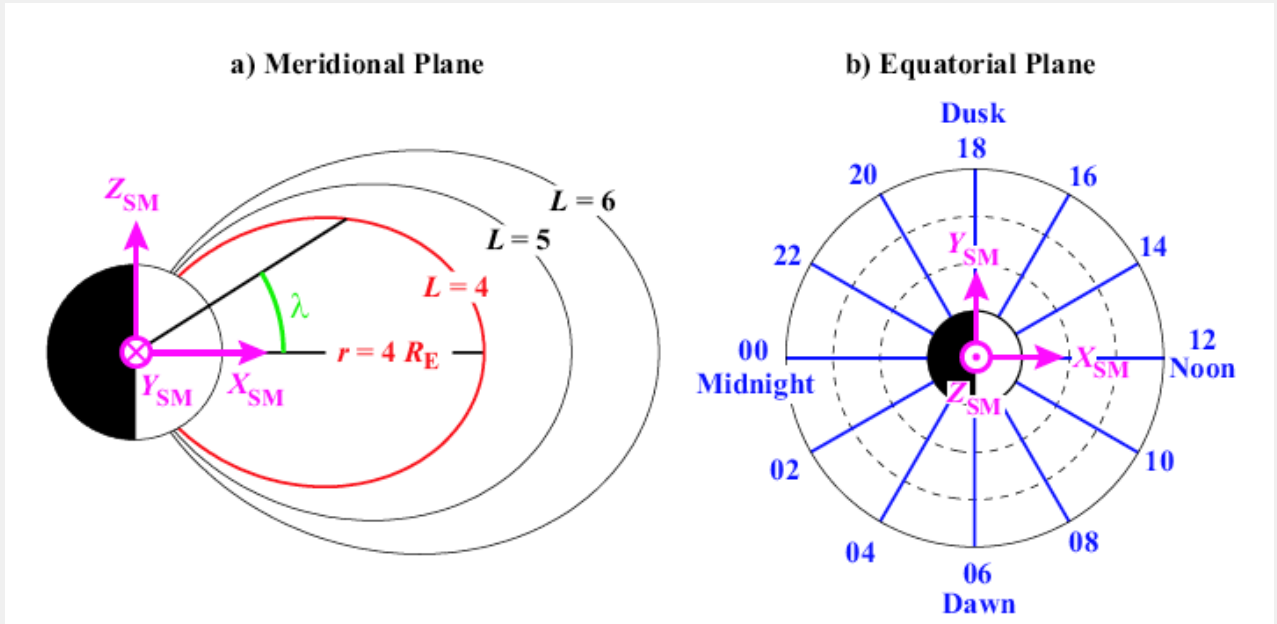
Plasmasphere

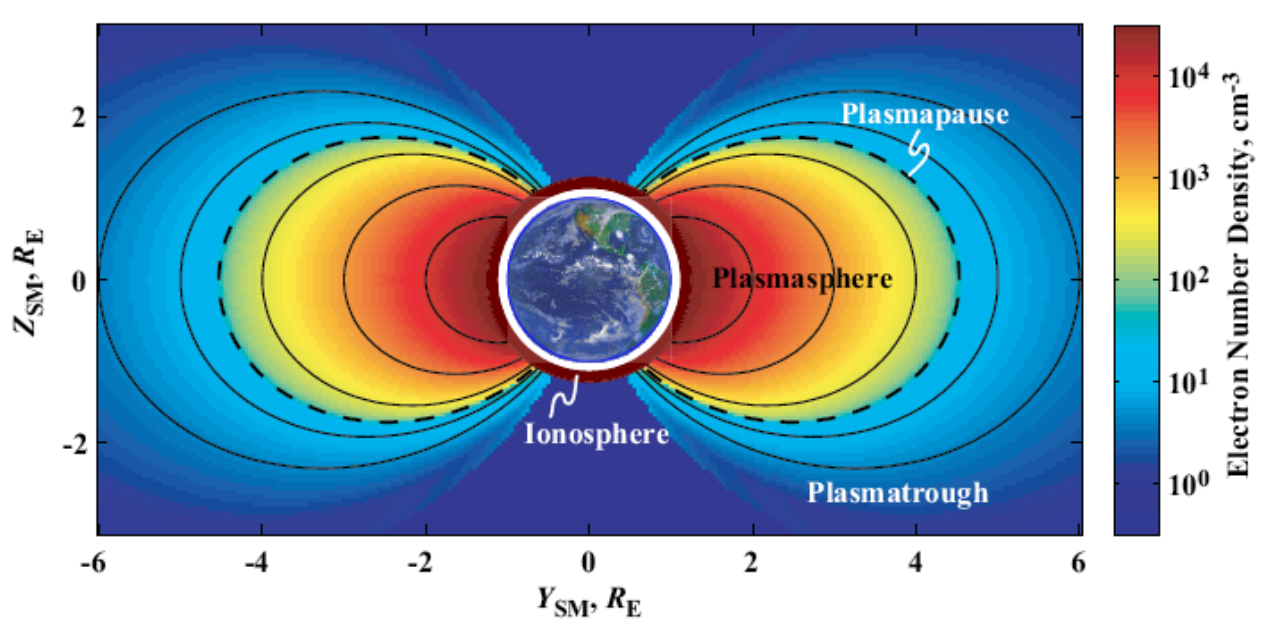
cold. 0.1 – 1 eV

Density $10 - 10^4\ cm^{-3}$

$\frac{B^2}{2\mu_0} > \frac{1}{2}\rho V_{th}^2$

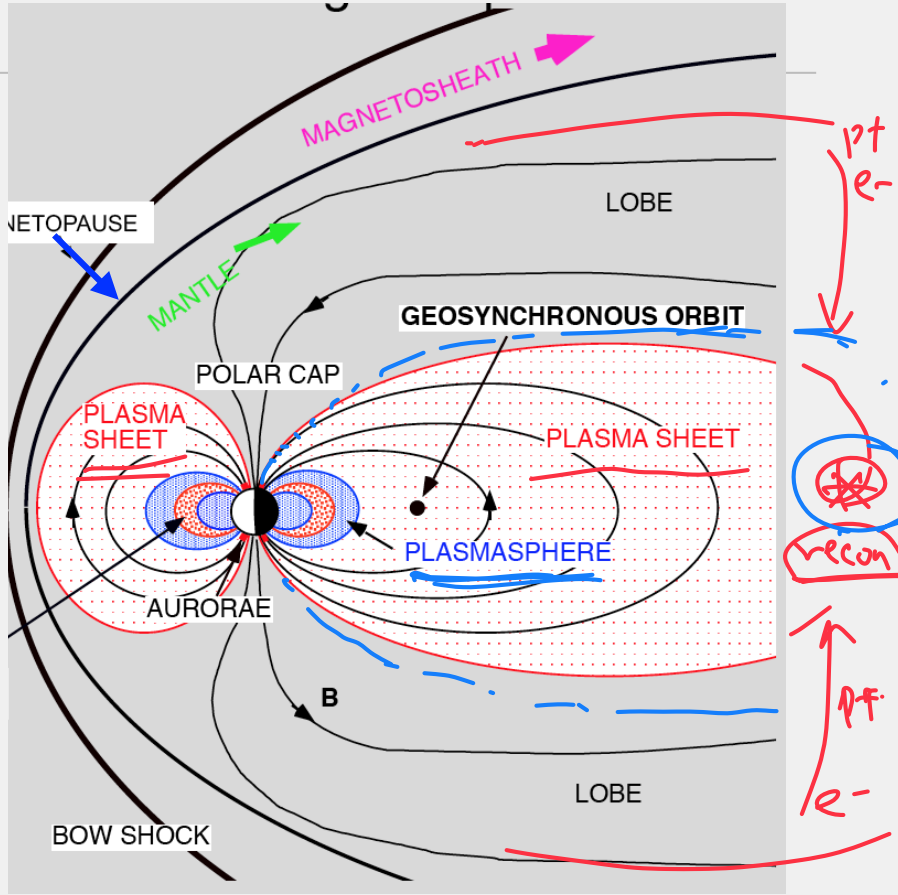






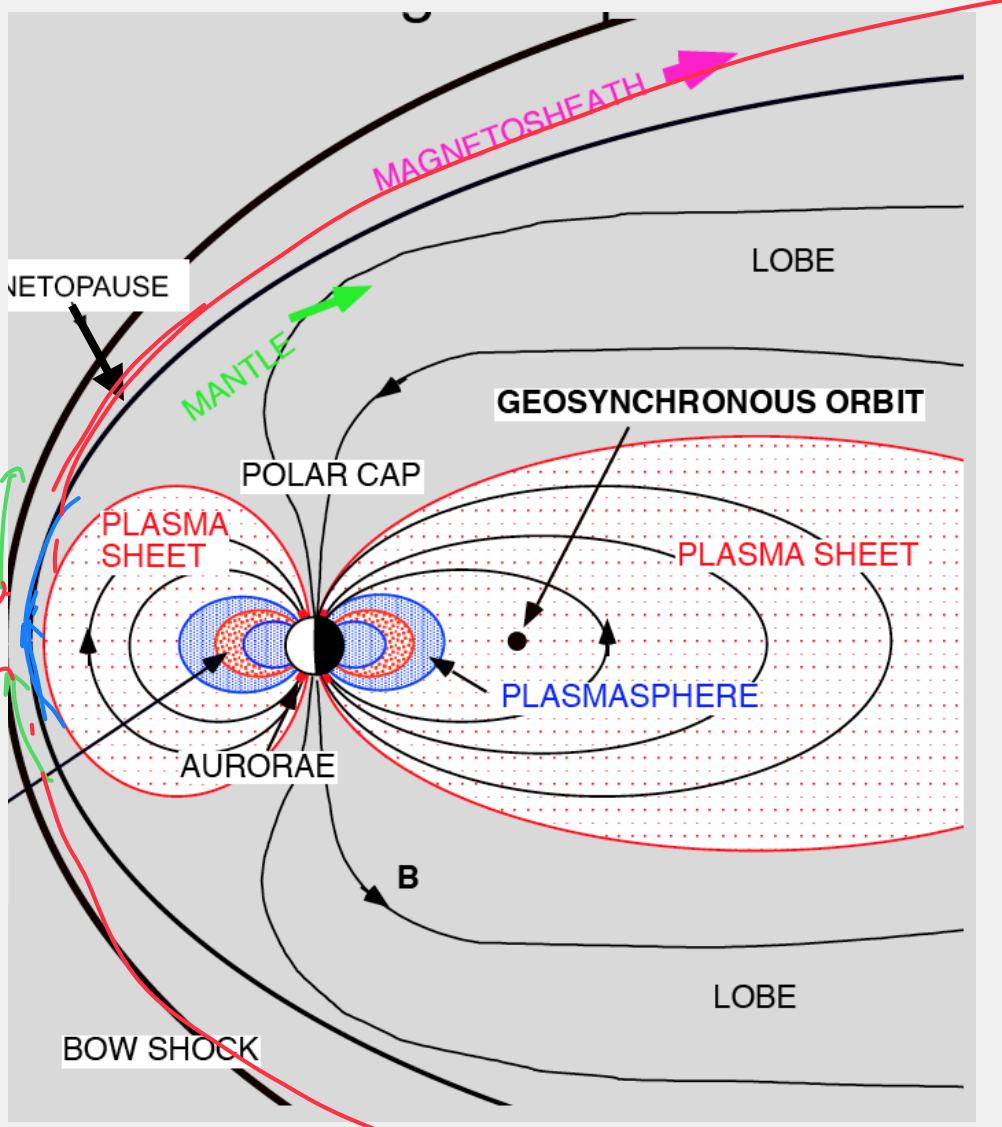
Plasma Sheet

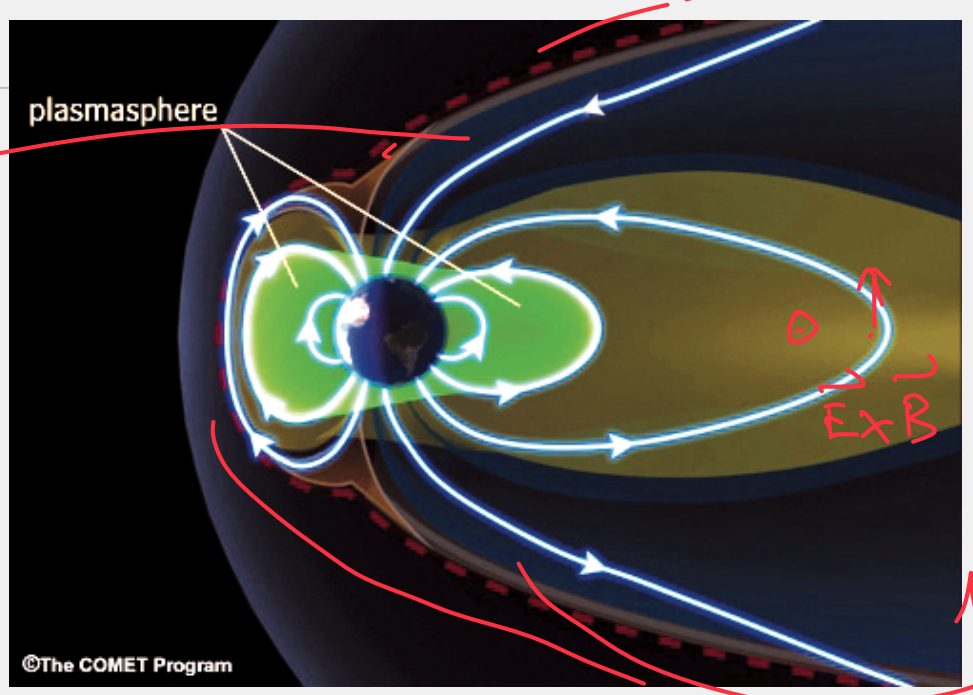
- The **plasma sheet** may be viewed as the extension of the **plasmasphere** beyond the **plasmapause**
- Plasmasphere partially corotates with the Earth's magnetic field; **plasma sheet does not** (field is too weak) – plasma "**beta**"
  - Plasmasheet electrons are **hot**: millions of Kelvins! (10 MK ~ 1-10 keV)
  - Densities are low: 0.1–1 particles/cm^3
    - Compare to plasmasphere: 100 particles/cm^3
  - Particles are mix of **solar wind** and **ionosphere outflow** (but energized)
  - Maps to Auroral region of Earth surface

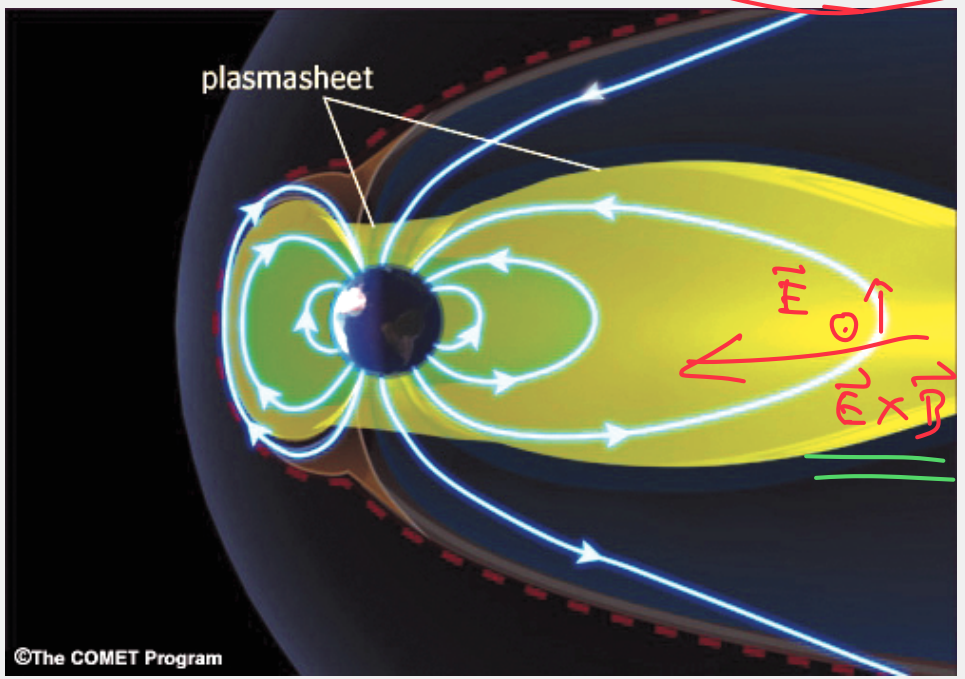


plasmasheet

Plasmasheet

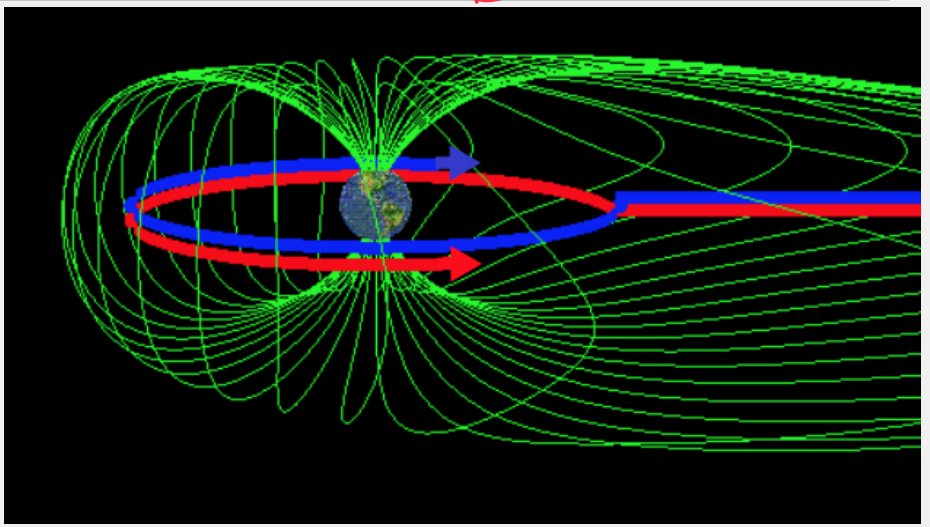


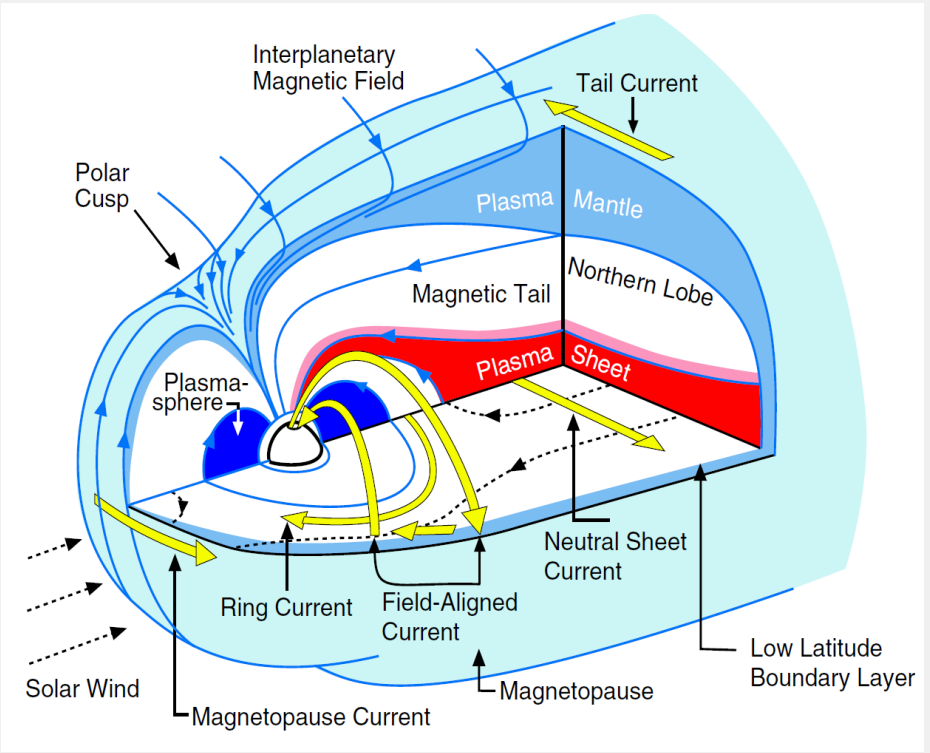




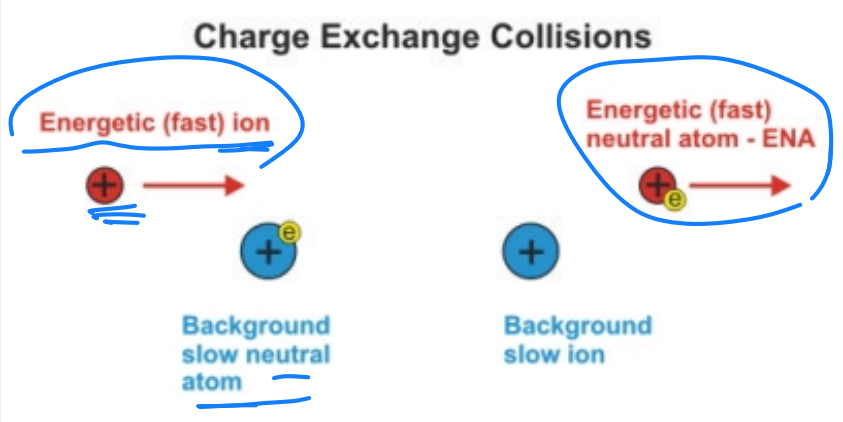
Ring Current

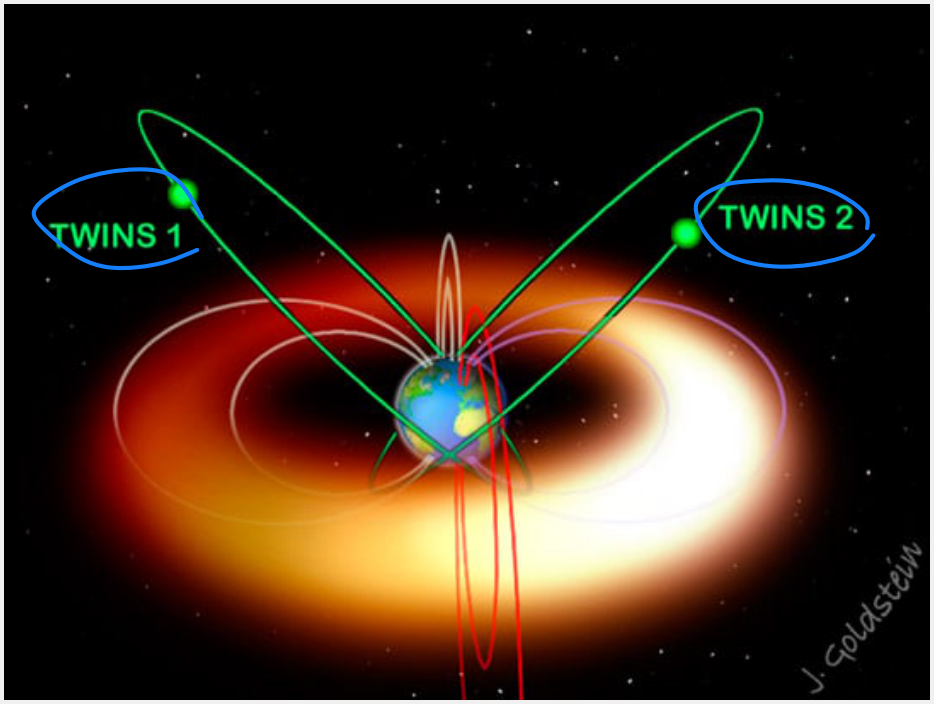
- Source: energetic plasma from the tail moving inward with magnetic field (**ExB** drift)
- Confined to 3–6 R_E
- Caused by the combination of the initial **ExB** drift (charge-independent), then Gradient and Curvature drifts (energy dependent); protons drift westward, electrons eastward.
- Average energy of ~80 keV; most from 10–250 keV
  - Current leads to a **reduction** in the magnetic field in the **inner** magnetosphere (compared to dipole field), and **enhancement** in the **outer** magnetosphere
- Typically 10^{-8} A/m^2 ; I ~ 1-3 MA



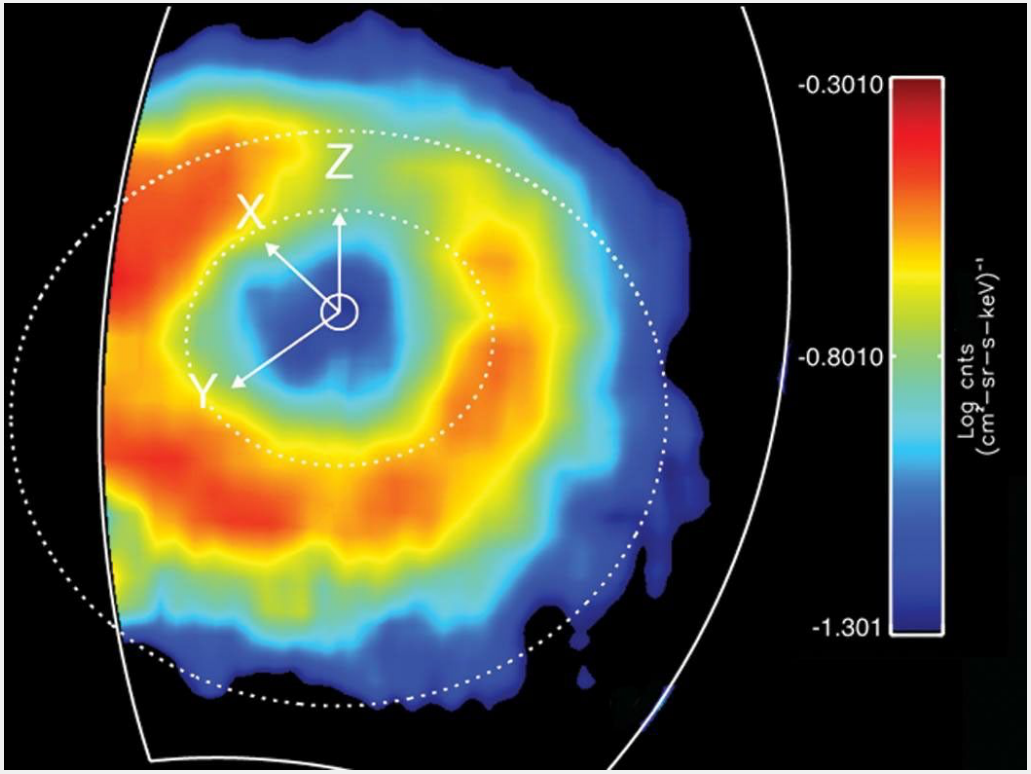


Ring Current does look like a ‘ring’ from ENA measurements





Earth’s Ring Current

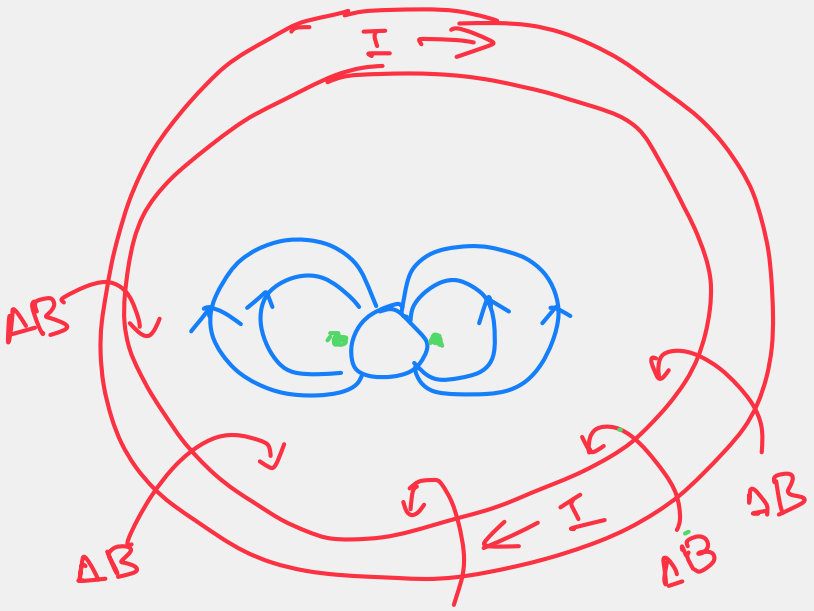


Saturn’s Ring Current

Why does ring Current reduce B field on surface?

Loop of current

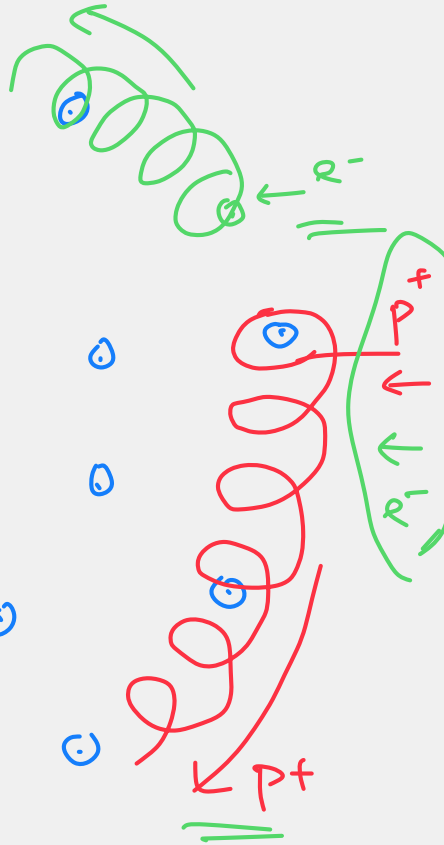
B_{rc} or $\Delta B = \frac{\mu_0 I}{2R}$



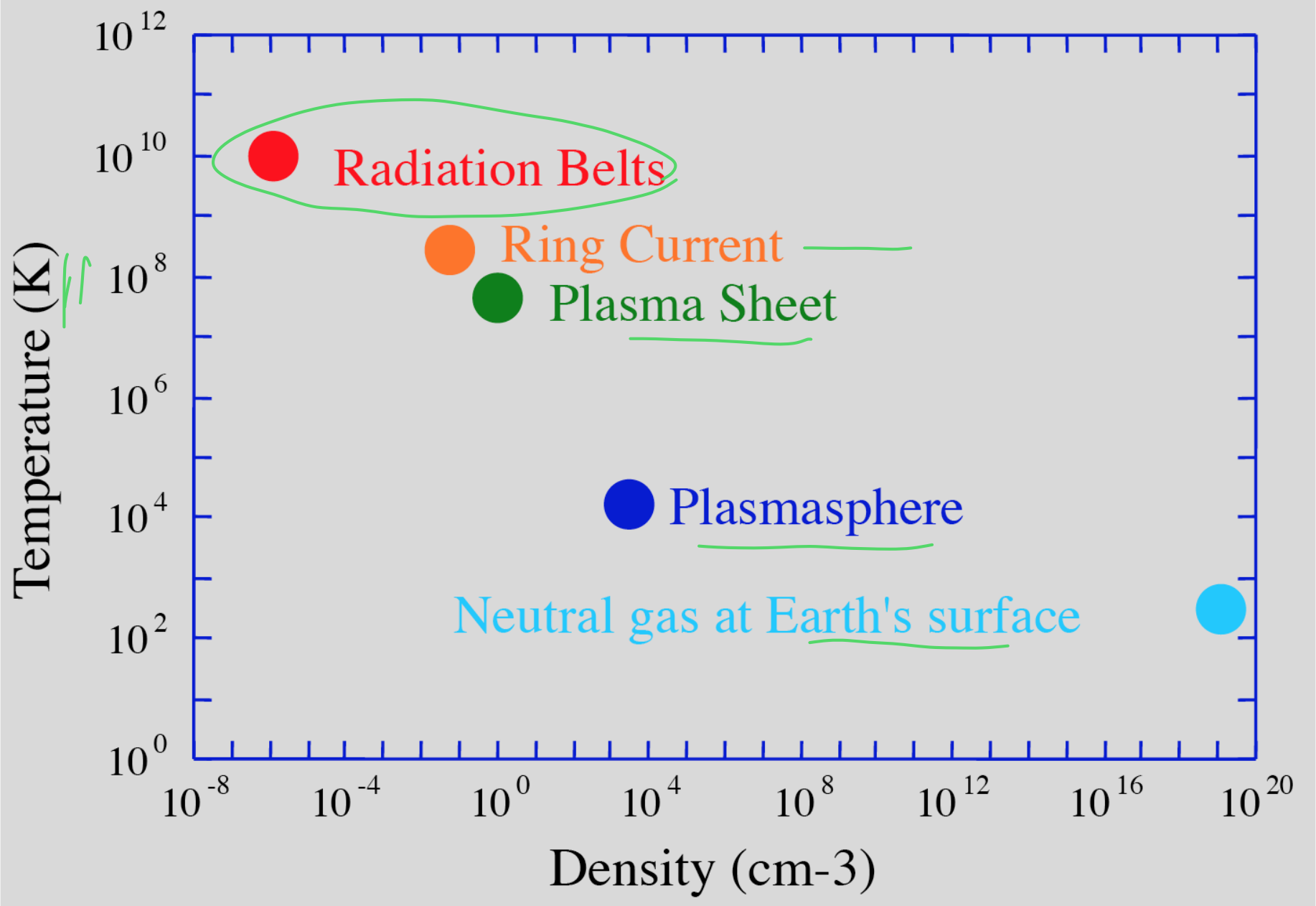
$B_E + \Delta B$

opposite direction of B_E at Earth's surface

look down from the North

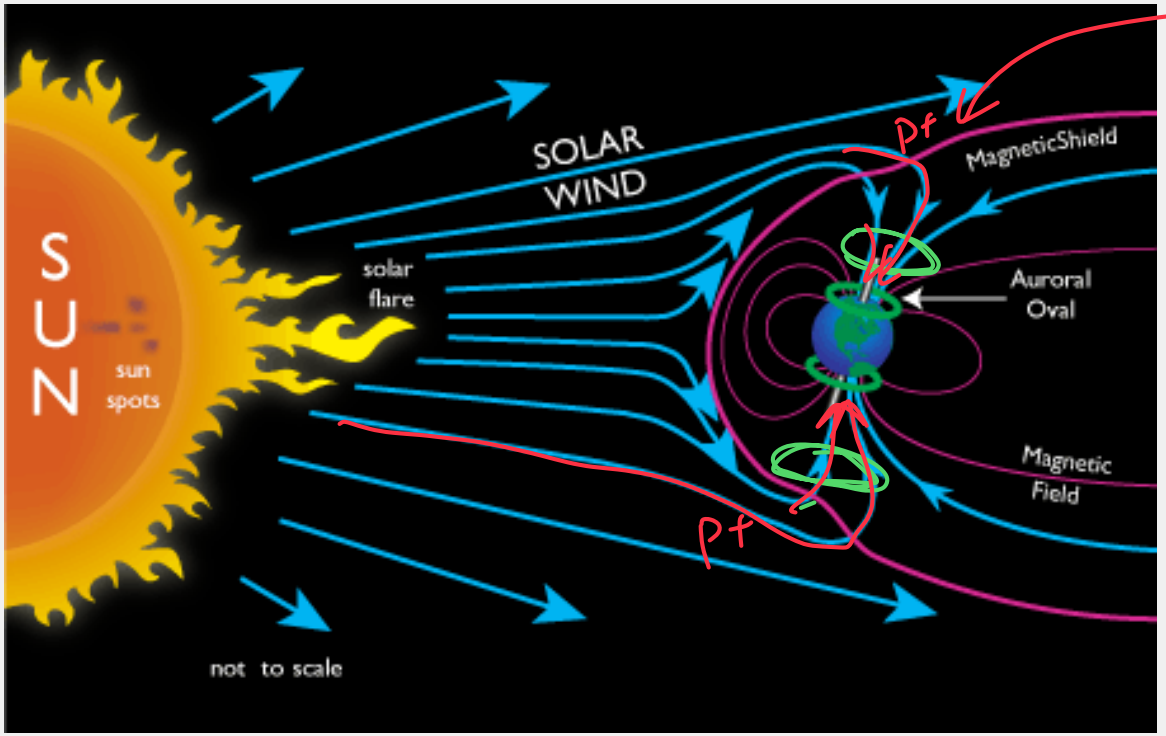


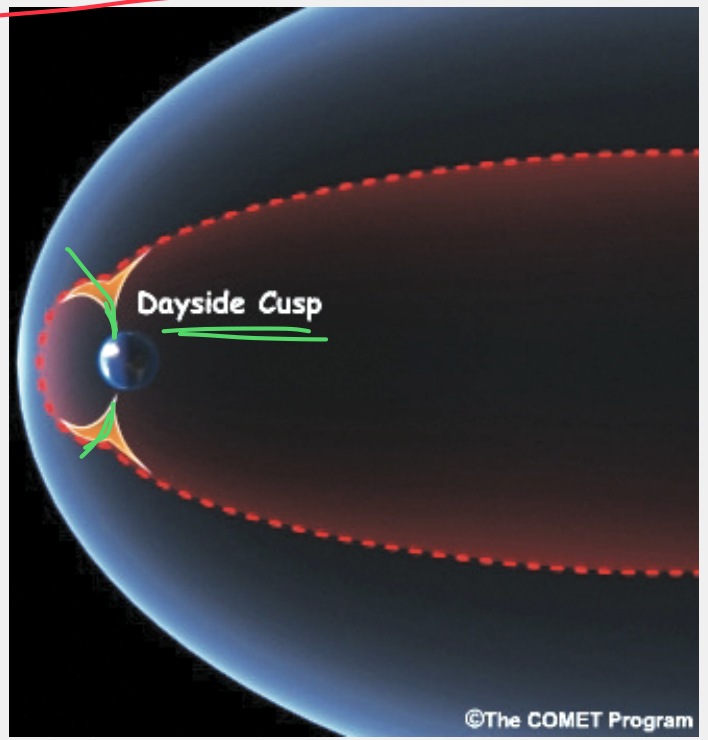
Plasma populations



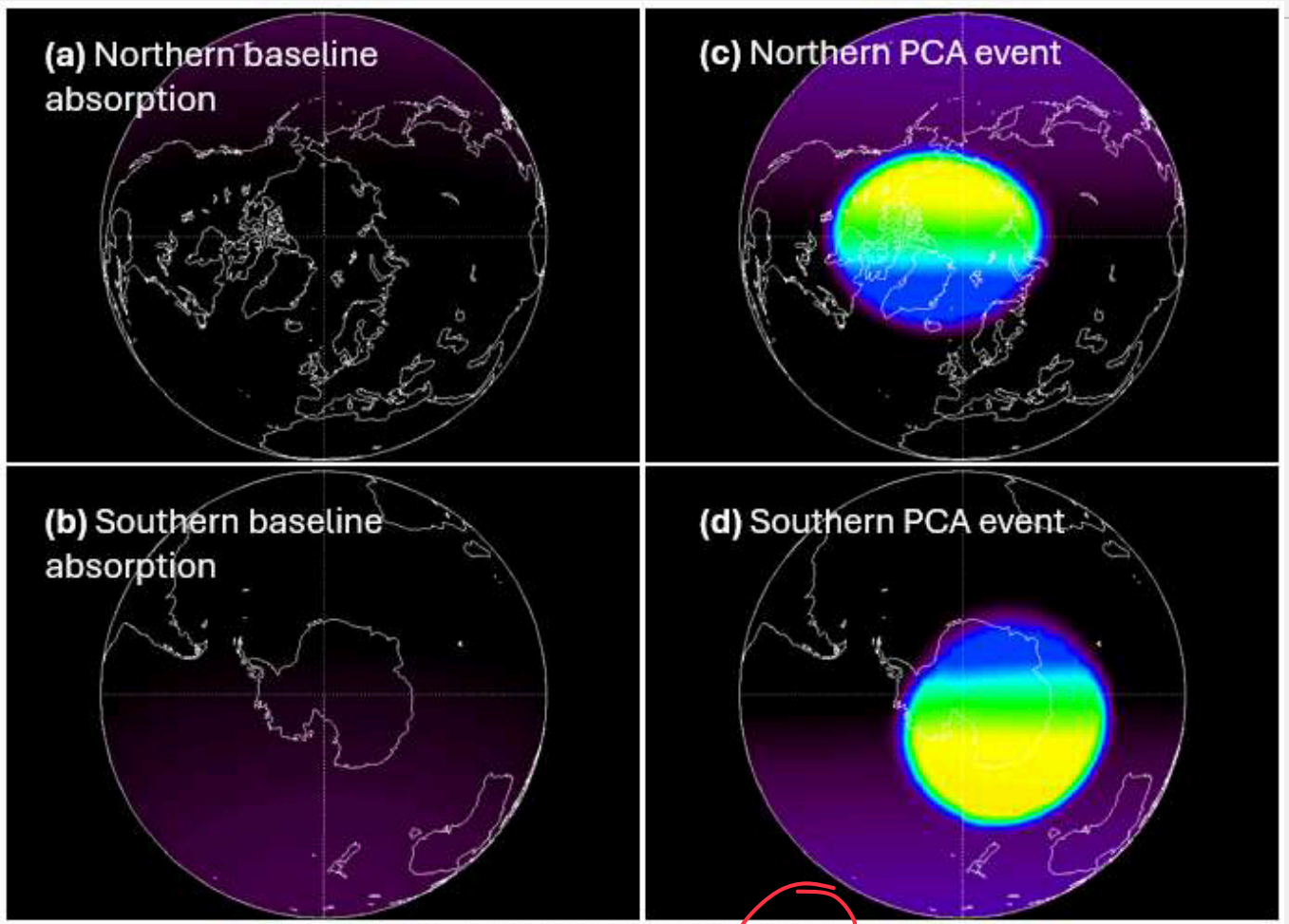
Polar Cap / Cusp Region

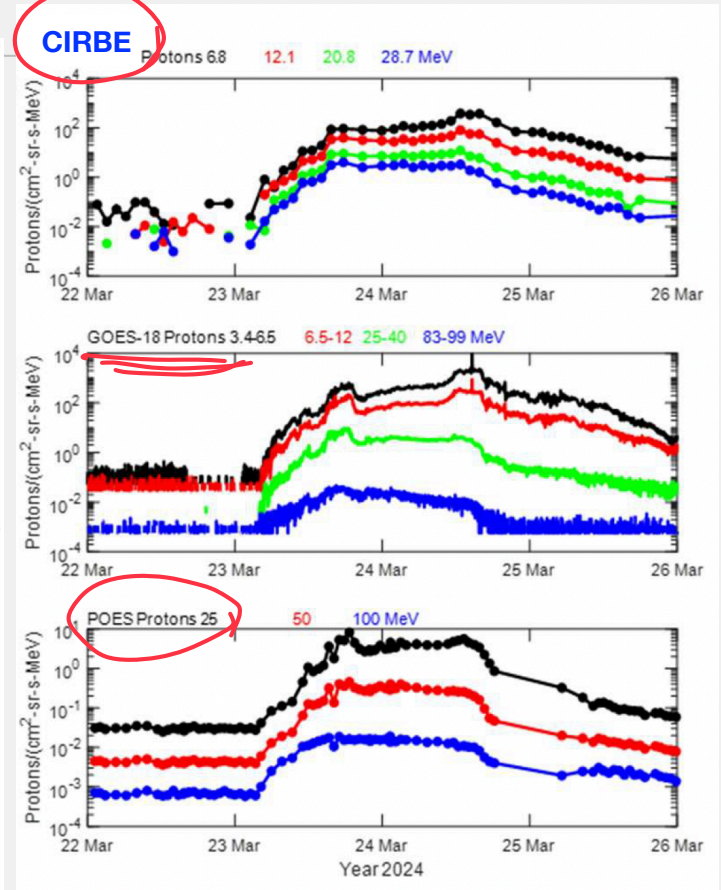
- Solar Energetic Protons (SEPs) and solar wind plasma can enter the magnetosphere directly in the cusp and polar cap region
- SEPs lead to energy deposition in D-region ionosphere => PCA (Polar Cap Absorption)
- Cusp is direct connection between atmosphere and magnetosheath
  - Also where ions can flow **outwards** from ionosphere into magnetosphere





Polar Cap Absorption (PCA)

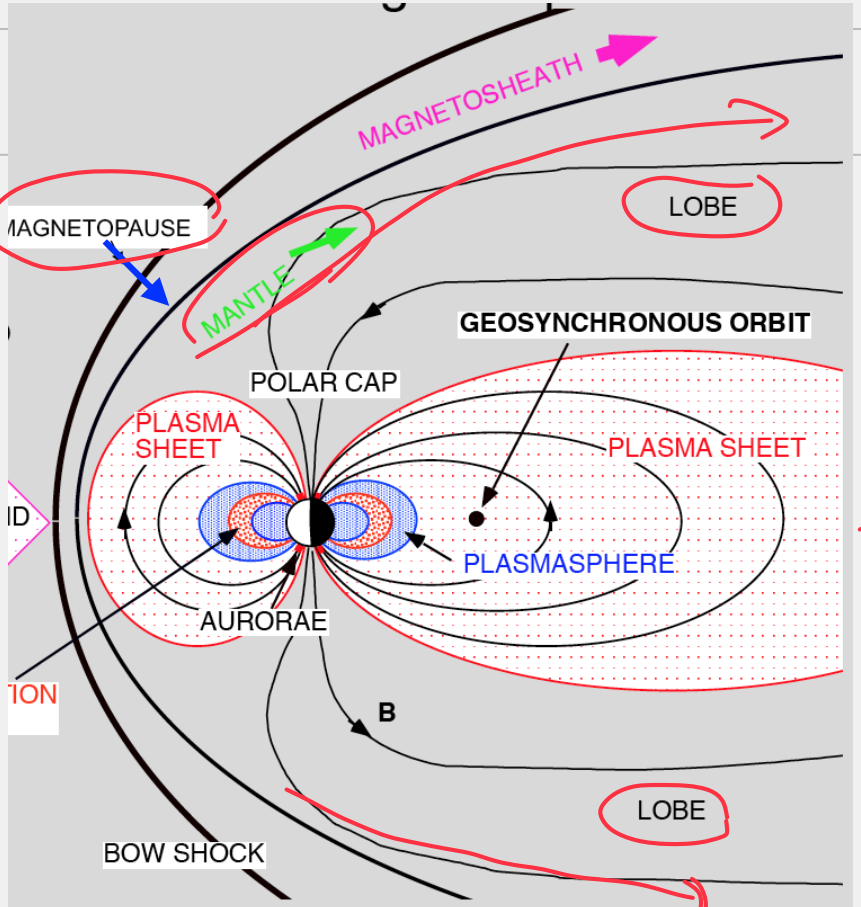


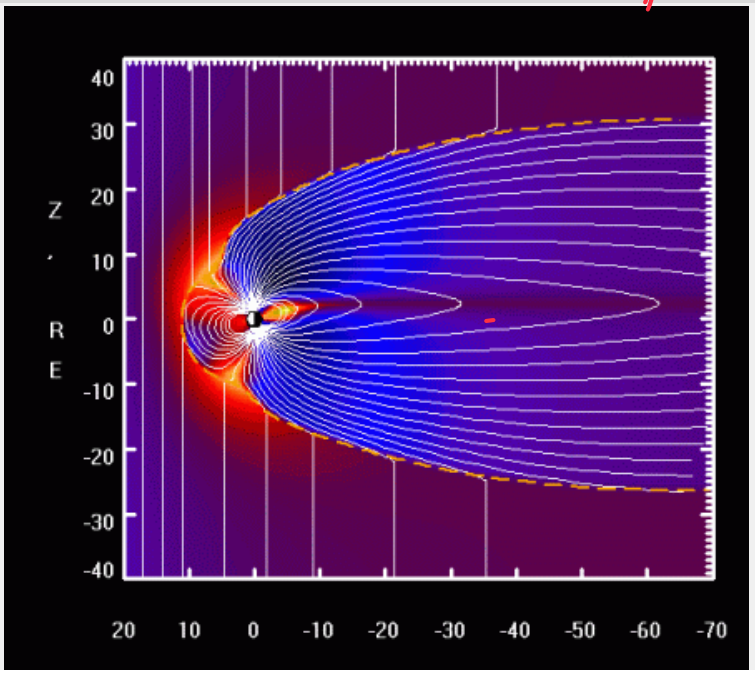


Sample output from NOAA's D-Region Absorption Prediction (D-RAP) model, with colors showing highest radio frequency affected by 10 dB absorption. Panels (a) and (b) show model results from 2023-11-03 00:08 UTC. This was a quiet day, with no precipitating SEPs or radio absorption. Panels (c) and (d) show model results from 2024-03-24 03 00:08 UTC, a period in which there was a very strong SEP event, see measurements by CIRBE, GOES, and POES, up-right corner

Other Regions of Interest

- **Plasma Mantle**: tailward of the cusp region, containing solar wind particles; magnetic field is strong, prevents solar wind from getting in.
- **Lobes**: inside the mantles; plasma in these regions originates from ionosphere outflow, with very low density; field lines from polar regions.
- **Magnetotail**: very weak B field; subject to "flapping" and turbulence due to solar wind.
  - Tail reconnection pushes balls of plasma toward Earth as field lines convect!





Summary of Magnetospheric Plasmas

Table 7-1. Thermal Plasma and Field Values in Earth's Magnetosphere. Here we list characteristics of the thermal plasmas and fields in Earth's magnetosphere.

	Density (n)	Electron Temperature (T_e)	Proton Temperature (T_i)	Magnetic Field	Comments
Plasmasphere	$>10^2$ particles/cm^3	10^4 K	0.1 eV – 1 eV	5000 nT	Cold plasma of terrestrial origin
Plasmasheet	0.1 –1.0 particles/cm^3	2×10^6 K to 2×10^7 K	Always higher by a factor of 3–5 than the electron temperature	9 nT in deep tail	Thickness of 4–6 R_E, forms into the ring current at 5–6 R_E from Earth
Plasmasheet boundary layer	0.1–1.0 particles/cm^3	2×10^6 K to 1×10^7 K	1×10^7 K to 5×10^7 K	20 nT–50 nT at 20 R_E	Region that maps to auroral zone, producing discrete arcs
Lobe	10^{-3}–10^{-2} particles/cm^3	$< 10^6$ K	$< 10^7$ K	Few nT	Lowest densities found in the magnetospheric cavity
Mantle	0.5–50 particles/cm^3	10^5 K–10^6 K	5×10^5 K to 8×10^6 K	10 nT–30 nT	Entry layer into the magnetosphere of magnetosheath plasma

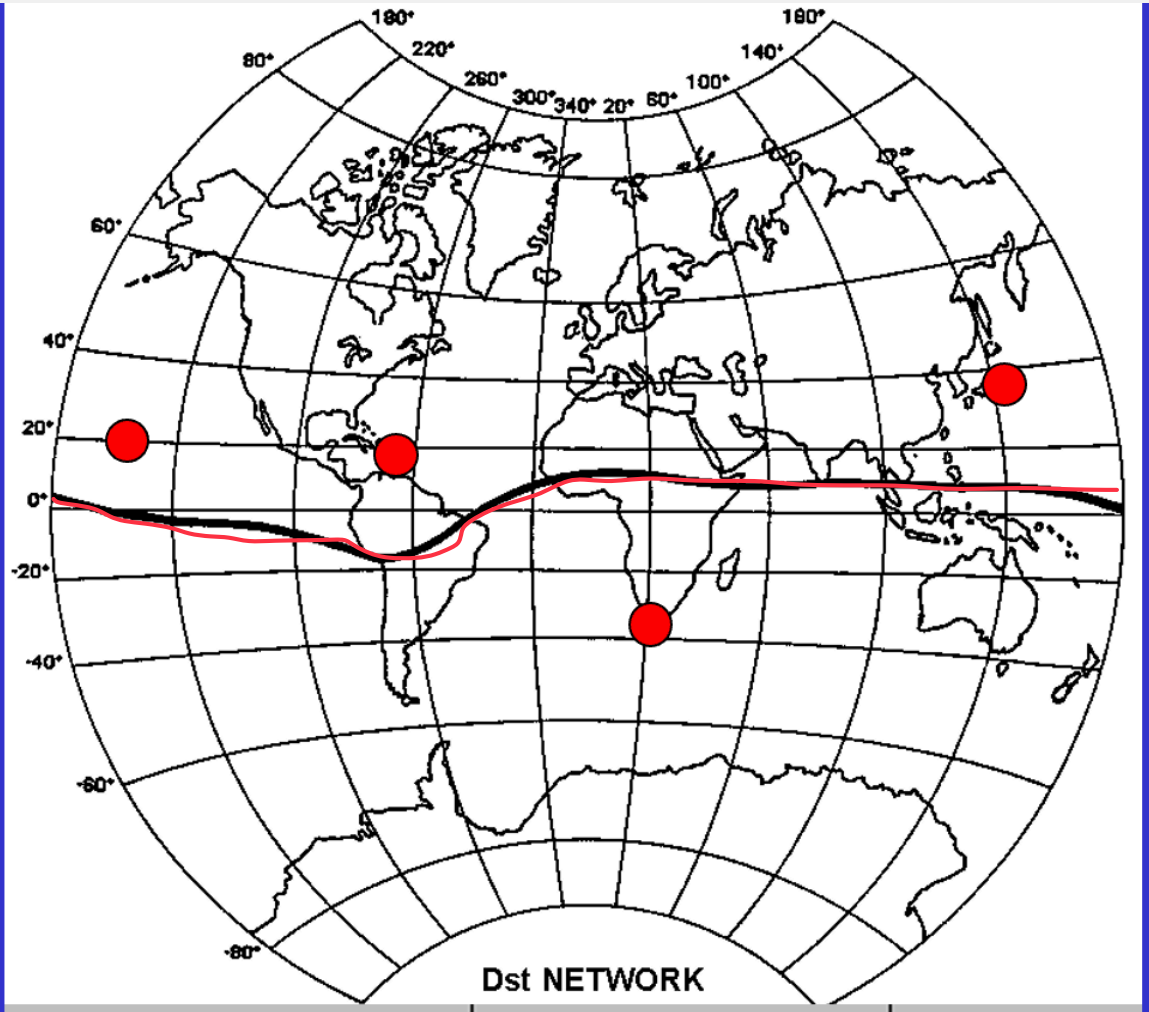
Geomagnetic Storm Progression

- **Initial phase: CME / CIR plasma structure arrives at magnetosphere**
  - Magnetosphere is compressed; a positive change in horizontal component at surface (Dst)
  - if IMF northward: no ring current enhancement
  - if IMF southward: energy enters magnetosphere and sudden storm commencement appears
- **Main phase: magnetosphere plasma and fields attempting to find new equilibrium**
  - Several hours to one day
  - Dst rapidly decreases to minimum
  - Ring current enhances and moves closer to Earth
- **Recovery phase: stored energy is dissipated**
  - ion loss to atmosphere and magnetopause; wave-particle interactions
  - Punctuated by impulsive substorms
  - days to several days long; may be followed by another storm! (Double Dst dip)

Geomagnetic Storm and Dst

How is the Dst index determined?

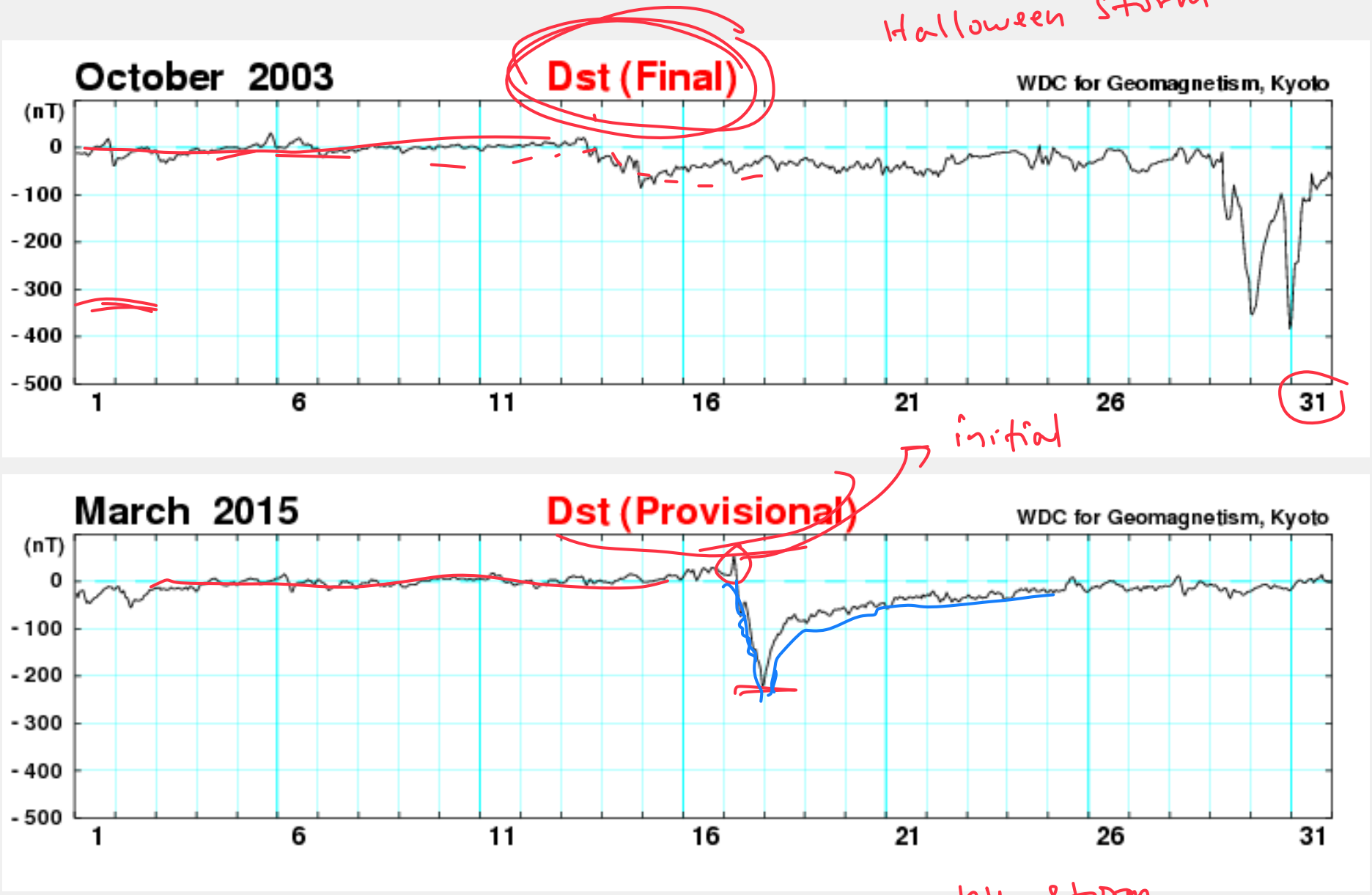
Ground Stations for the Dst Index



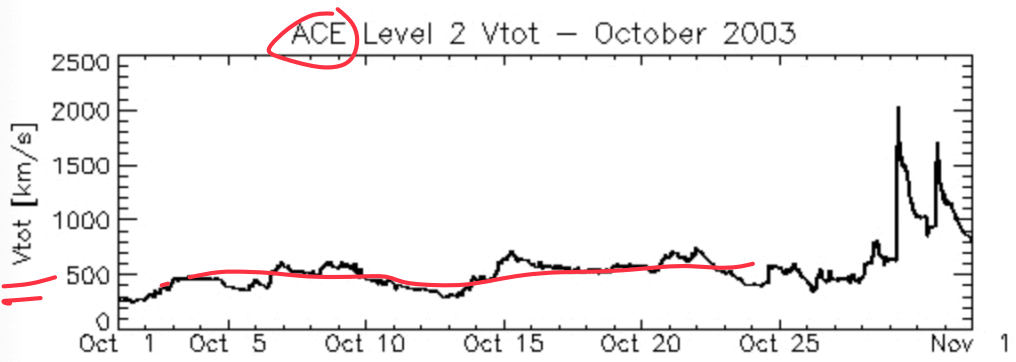
Names and Coordinates of the Ground Stations

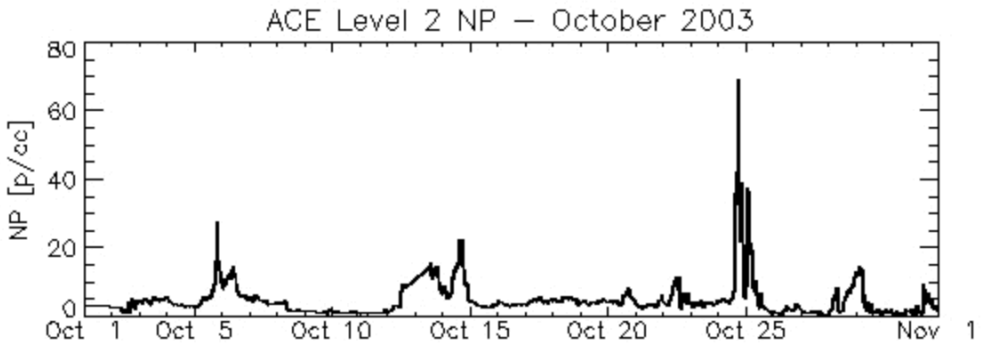
Observatory		Geographic		Geomagnetic
		Longitude (E)	Latitude	Dipole latitude
Hermanus		19.22°	-34.40°	-33.3°
Kakioka		140.18°	36.23°	26.0°
Honolulu	to April 1960	201.90°	21.30°	21.0°
	after April 1960	201.98°	21.32°	21.1°
San Juan	to January 1965	293.88°	18.38°	29.9°

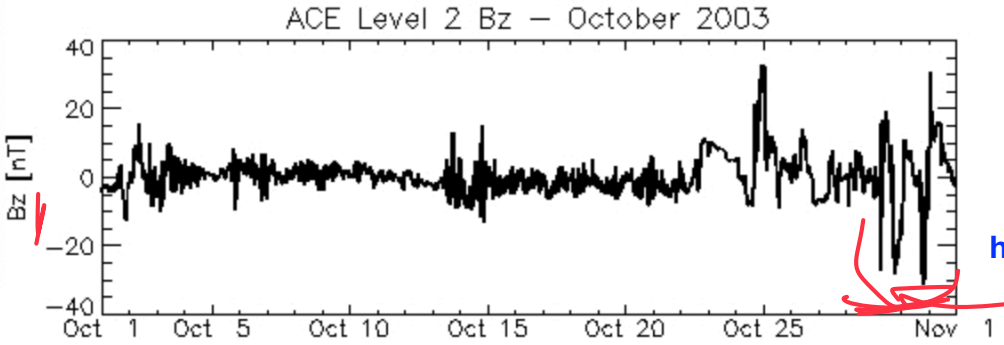
Geomagnetic Storm and Dst



Geomagnetic Storm and Dst Prediction



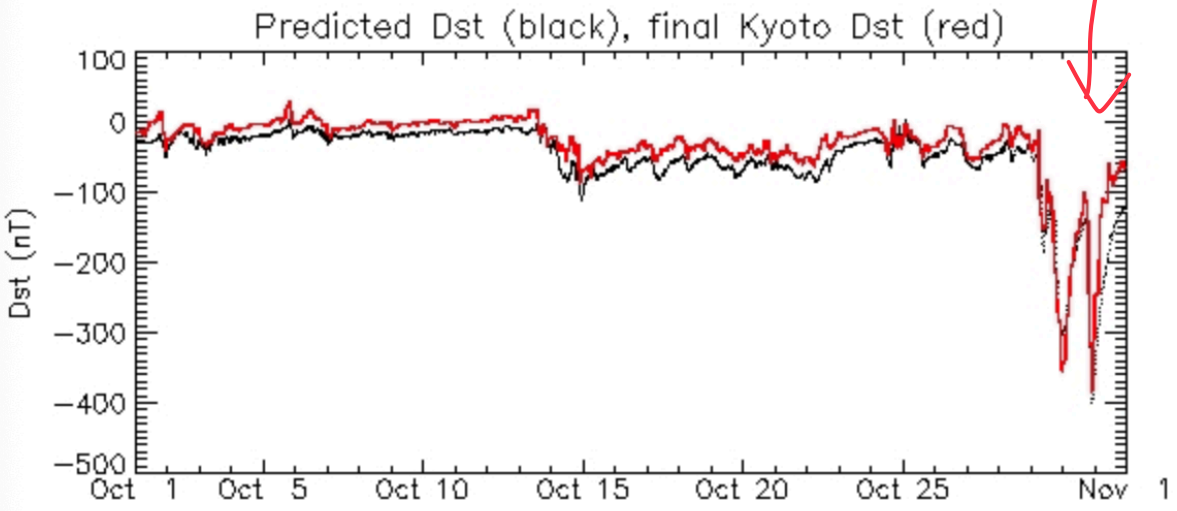




(Temerin and Li, JGR, 2002; 2006)

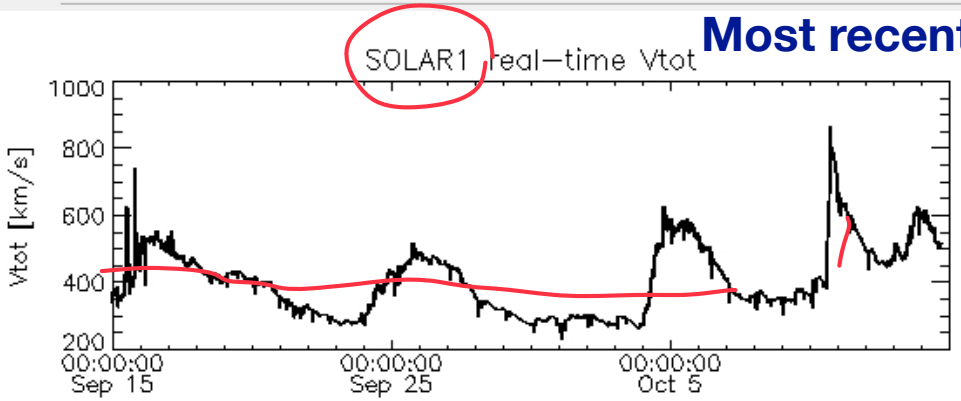
https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html

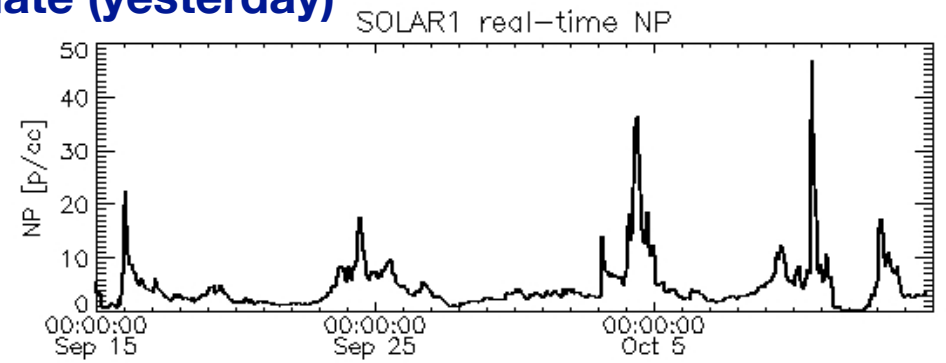
How is this possible?

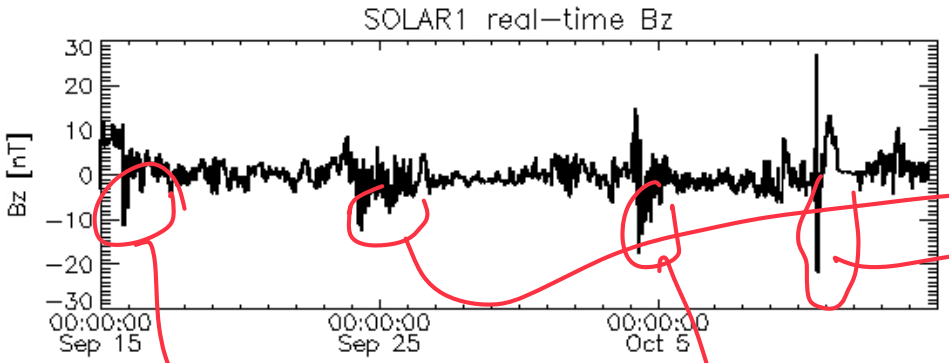


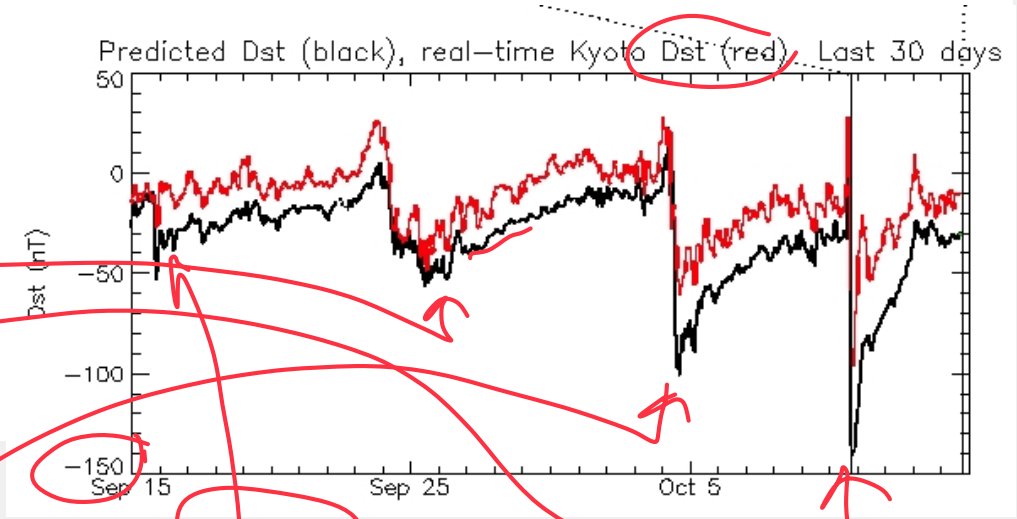
Geomagnetic Storm and Dst Prediction

Most recent update (yesterday)









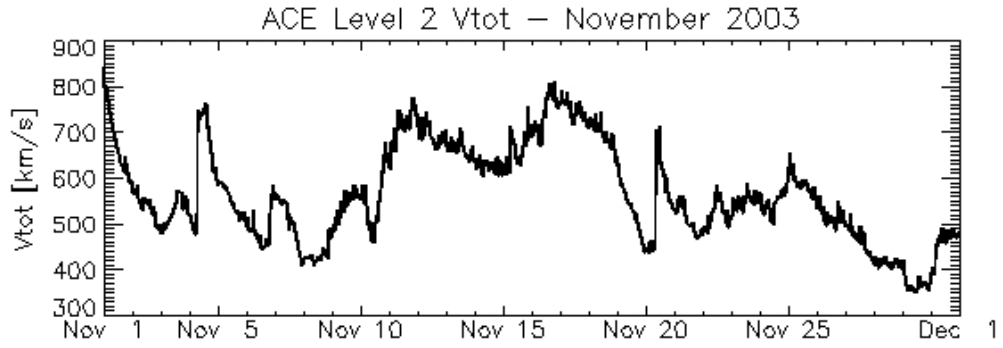
How is this possible?

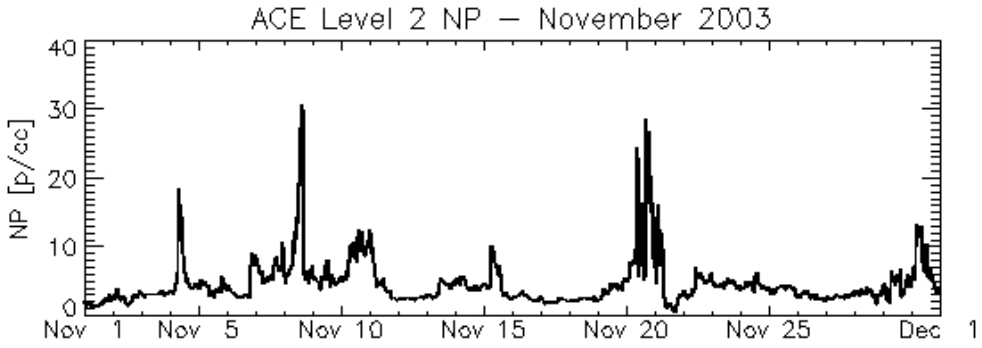
(Temerin and Li, JGR, 2002; 2006)

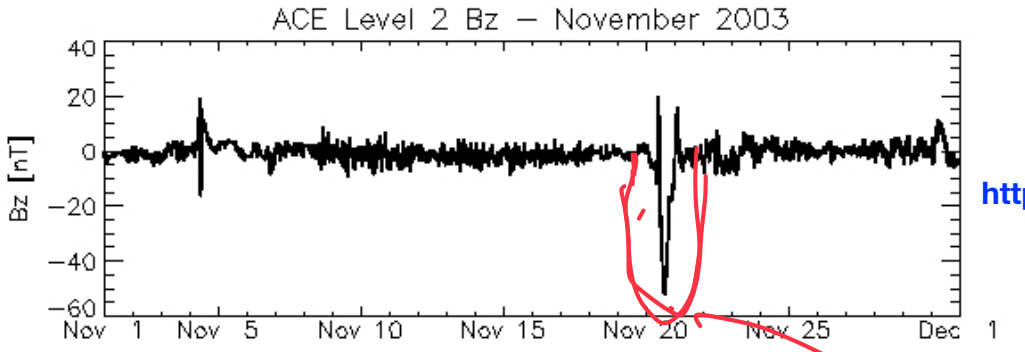
https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html

$$Dst = dst1* + dst2* + dst3* + (pressure\ term)* + (direct\ Bz\ term)* + (offset\ term).$$

The terms *dst*1, *dst*2, *dst*3, (*pressure term*), and (*direct Bz term*) were in our previous model. The terms *dst*1*, *dst*2*, *dst*3*, (*pressure term*)*, (*direct Bz term*)* differ, in part, in



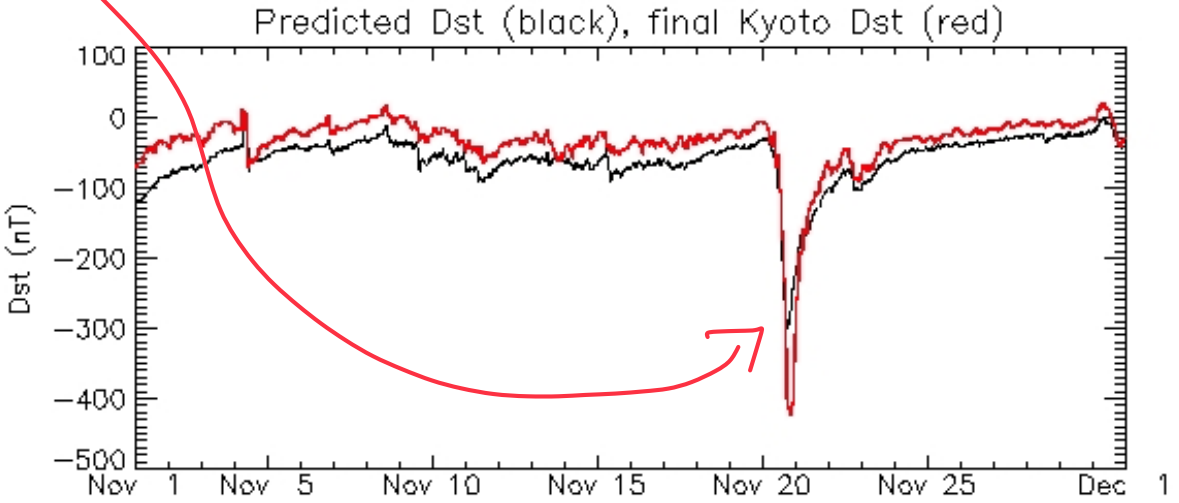




https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html

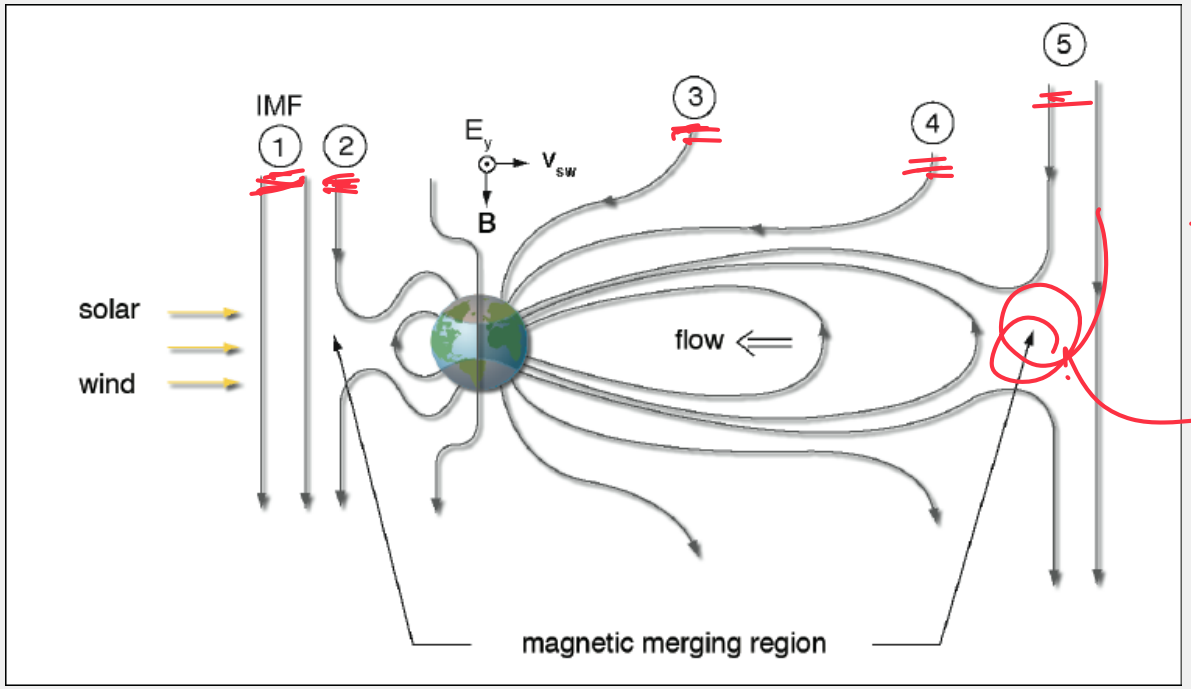
(Temerin and Li, JGR, 2002; 2006)





IMF Polarity and Reconnection

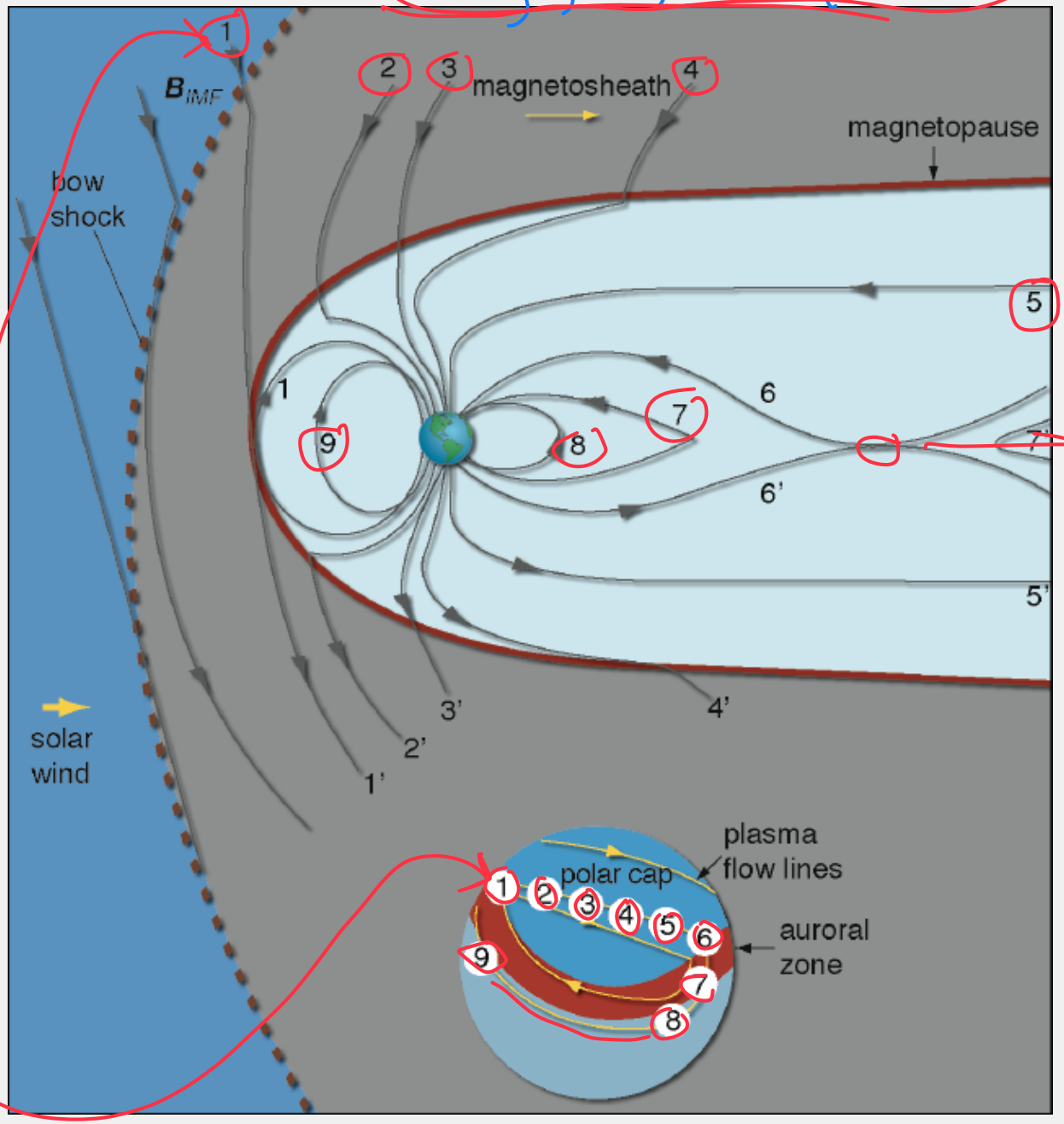
- IMF is "normally" pointing any direction, fluctuating
  - Approximately doubles B-field at magnetopause; hence factor of two in our calculation of standoff distance
- During CMEs, IMF at Earth can point south or north or any other direction for sustained time with a bigger magnitude
- Earth's magnetic field is pointing north…





Reconnection

- Solar wind carries plasma!
- At reconnection, that plasma can flow along field lines into the polar cap
- As field lines wrap around to the tail, polar cap signature draws a "loop"
  - Points 7, 8, 9 stick out from the page; field lines looping around dusk side of Earth

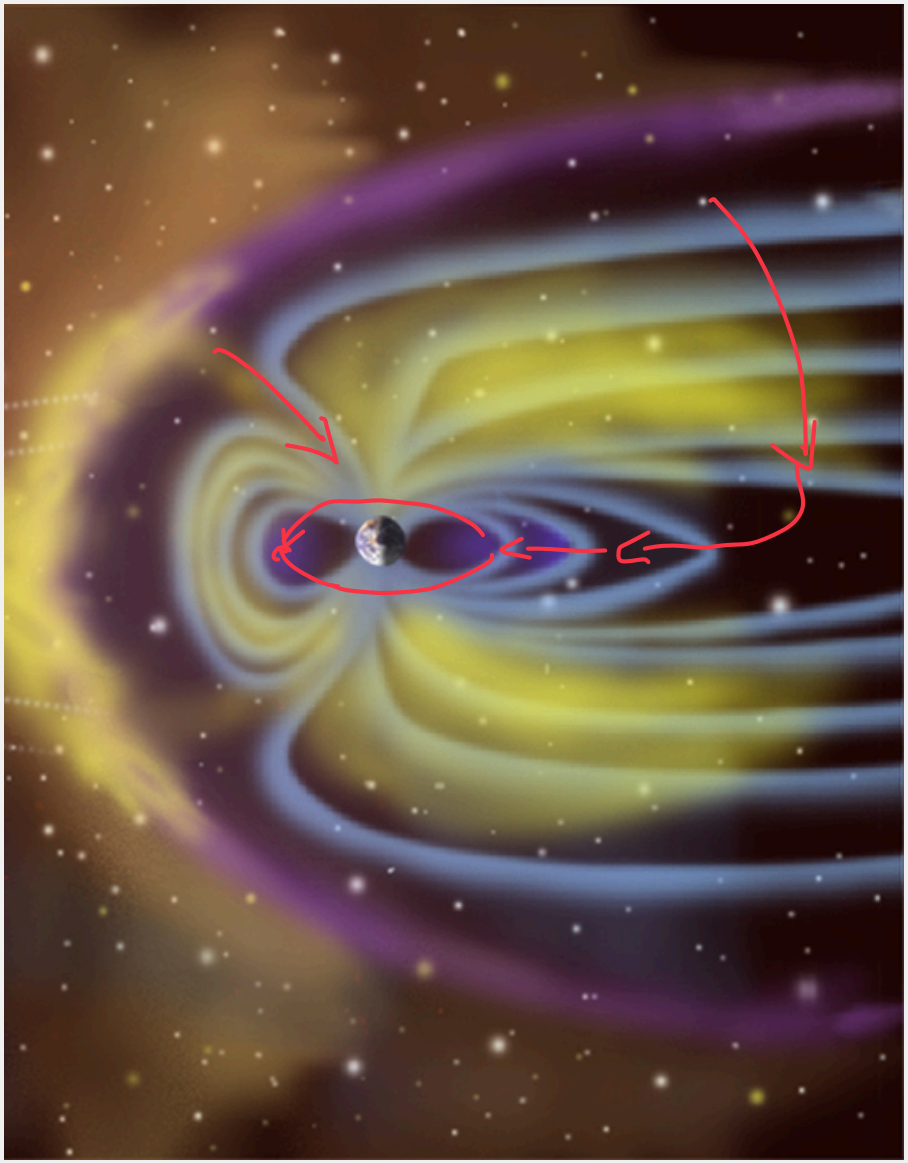


Geomagnetic Storms and Auroras



Storm Effects

- **What happens to the space environment during a geomagnetic storm?**
  - Magnetosphere gets compressed; magnetopause moves inward
  - Reconnection and field line convection
  - Solar wind / CME plasma gets injected into magnetosphere
  - Ring current gets enhanced 10x or more
    - Ring current enhancement leads to reduction of B-field on the surface by 1–2% within hours; recovery over 1–2 days
  - Plasmasphere shrinks (being eroded): plasmapause moves inwards
  - Radiation belts get enhanced: 10-100x more high-energy electrons and protons
  - Auroras occur; larger storm, auroras reach to lower latitudes



Example: March 1989 Event

